

Corso: Large Language Models

Quiz: Exam (2025-02-11)



PORTALE
ESAMI

Iniziato	11 febbraio 2025, 14:11
Stato	Completato
Terminato	11 febbraio 2025, 15:26
Tempo impiegato	1 ora 15 min.
Valutazione	10,38 su un massimo di 18,00 (58%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12

Domanda 1

Completo

Punteggio
ottenuto 0,00
su 2,00

2 points (no penalty for a wrong answer)

What is the likely output of the following code snippet?

```
1 from transformers import AutoModel, AutoTokenizer
2
3 model_name = "bert-base-uncased"
4 model = AutoModel.from_pretrained(model_name)
5 tokenizer = AutoTokenizer.from_pretrained(model_name)
6
7 prompt = "The result of the operation 2 + 2 is "
8 tokens = tokenizer(prompt, return_tensors="pt")
9
10 result = model.generate(**tokens)
11 print(tokenizer.batch_decode(result))
12
```

Provide a brief explanation (max 50 words).

If an error occurs, indicate the line number and briefly explain why.

the output of this snippet code is the attention and the mask of the string prompt.
we open the pretrained model using the command from_pretrained, we open also the
tokenizer specialized
for this typology of task, then the output result will contain the tokenize version of
the string, using
the command generate we define the generated token.
the last command convert the result in the decode token, so from token to label(word)
t

Numero parole: 73, superiore al limite massimo di 50 parole

The code produces an error because AutoModel does not support .generate().
Additionally, BERT is an encoder-only model, so the code is also conceptually incorrect.

Commento:

The code generates an error: BERT is an encoder-only model. As such, it is not built to produce new tokens autoregressively (as .generate() does).

Domanda 2

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

What is the purpose of reflection in an agentic system?

- ☐ a. To eliminate the need for structured workflows
- ☐ b. To generate human-like responses using predefined scripts
- ☐ c. To store knowledge permanently in short-term memory
- ☐ d. To execute tasks without modifying strategies
- ☒ e. To evaluate past decisions and improve future performance ✓

Risposta corretta.

La risposta corretta è:

To evaluate past decisions and improve future performance

Domanda 3

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for a wrong answer)

What is the main difference between encoder-decoder cross-attention and encoder self-attention?

- ☐ a. In encoder-decoder cross-attention, the decoder attends only to the current token in the input sequence, while in encoder self-attention, the encoder attends to all tokens in the input
- ☐ b. Encoder self-attention attends to the encoder's output, while encoder-decoder cross-attention attends to the decoder's previous tokens
- ☐ c. In encoder self-attention, the decoder attends to the input sequence, whereas in encoder-decoder cross-attention, the encoder attends to the output sequence
- ☐ d. Encoder self-attention is used in the decoder, and encoder-decoder cross-attention is used in the encoder
- ☐ e. None of the other options is correct
- ☐ f. Encoder-decoder cross-attention attends only to the previous tokens in the output sequence, while encoder self-attention attends only to tokens in the input sequence
- ☒ g. Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output ✓

Your answer is correct.

La risposta corretta è:

Encoder self-attention only uses information from the input sequence, while encoder-decoder cross-attention allows the decoder to attend to the encoder's output

Domanda 4

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

You are given the following sequence of characters.

a c e a c e a a a a c b

Initially, each character is a separate token.

Apply **Byte-Pair Encoding** to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

1) if the merged tokens are aa and b, write: **aa,b**

2) if the merged tokens are x and y, write: **x,y**

1st merge: a,a ✓

2nd merge: a,c ✓

3rd merge: ac,e ✓

Domanda 5

Risposta
errata

Punteggio
ottenuto -0,25
su 1,00

1 point (-25% penalty for a wrong answer)

Which of the following best describes how BERTScore evaluates text similarity?

- ☐ a. It measures sequence similarity using n-gram co-occurrence weighted by IDF scores
- ☐ b. It uses a fine-tuned BERT model to classify whether the reference and candidate texts are semantically related or not
- ☒ c. It predicts a similarity score by treating the evaluation task as a masked language modeling problem, where BERT predicts the score as the next token ✗
- ☐ d. It assigns similarity scores based on syntactic parse trees rather than token embeddings
- ☐ e. It uses cosine similarity on static word embeddings trained with Word2Vec or GloVe
- ☐ f. It applies a bag-of-words model to compare token frequency distributions between reference and candidate texts
- ☐ g. It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores

Your answer is incorrect.

La risposta corretta è:

It computes token-level similarity by aligning contextualized embeddings from a pre-trained transformer model and aggregating scores

Domanda 6

Risposta
corretta

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for a wrong answer)

Apply absmax quantization to the following values x1, x2, x3, x4, to map them to int8.

- x1 = -5
- x2 = 4
- x3 = -1
- x4 = 0

Write the quantized versions in the following boxes.

quantized(x1) = -127 ✓

quantized(x2) = 102 ✓

quantized(x3) = -25 ✓

quantized(x4) = 0 ✓

Domanda 7

Parzialmente
corretta

Punteggio
ottenuto 0,50
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following are benefits of using open-source datasets for training LLMs?

- ☒ a. Authenticity and credibility of real-world data ✓
- ☐ b. Guaranteed high accuracy in model predictions
- ☐ c. Ability to acquire timely data and share data freely
- ☐ d. Elimination of bias in training data
- ☐ e. LLMs no longer require fine-tuning

Risposta parzialmente esatta.

Hai selezionato correttamente 1.

Le risposte corrette sono:

Authenticity and credibility of real-world data,

Ability to acquire timely data and share data freely

Domanda 8

Completo

Punteggio
ottenuto 2,00
su 3,00

3 points (no penalty for wrong answers)

Consider a scenario where you have:

- A natural language description of a function's requirements (REQS)
- A set of test cases (TESTS) specifying expected inputs and outputs

You need to design an agentic architecture using LLAMA-based agents to:

- Generate an initial implementation of the function based on REQs
- Refine the code iteratively until all test cases pass

Describe the architecture of LLAMA agents that perform these steps. For each agent:

- Define its role in the pipeline
- Specify its prompt, including
 - System message: Defines the agent's behavior
 - Context: Information given to the agent from previous steps
 - Possible inputs: Data received from prior agents
 - Expected output: The structured result the agent should return

Notice: It is not necessary to write Python calls to transformers, code for output parsing or actual code for test execution and parsing. Focus on the agentic workflow and prompt engineering techniques.

we have:

```
requirements = " big string with our requirements "
```

```
list of sets= [ test1 , test2 , test3 ]
```

AGENT : ROUTER to generate our requirements

CONTEXT: first of all, we have to extract our requirements, so we can use an router agent with a prompt few shot, to first extract the requirements

Input:

prompt=

" you are the best router in the world, that you are able to extract requirements,this will be its structure :

1. define the requirements
2. default branch

#example1

input : here there is an example

output: here there is an example

#example2

input: here there is an example

output; here there is an example

Now, its your turn, given this requirements extract the data:

Input: { input string }

Output:

"

Output:

1) answer

2) default branch

AGENT: to tweak our test merge with chain of thought with few shot prompting

CONTEXT: prompt used chain of thought iteratively to tweak the test, and check the validity using a test (like mutation analysis or test coverage)

for loop:

INPUT:

previous answer : test

prompt = " you are an agent able to tweak the test :

u have some example:

#example 1: test

output: output test

now its your turn:

input: { previous test }

output:

"

OUTPUT: The tweaked test

an if condition to check if all test pass:

break if the test pass,

or continue iteratively to tweak the test

Commento: the objective of the second agent would not be to tweak the tests but to refine the code to pass a set of stable test cases

2 points (no penalty for wrong answers)

Consider the following cosine similarity table. Which of the following threshold is more dependable in terms of TPR and FPR?

Domanda 9

Completo

Punteggio
ottenuto 1,00
su 2,00

T = 0.70, 0.80, 0.90

	Exp1	Exp2	Exp3	Exp4
Act1	0,71	0,65	0,62	0,30
Act2	0,71	0,74	0,30	0,20
Act3	0,54	0,05	0,60	0,55
Act4	0,80	0,40	0,15	0,92

1)

T= 0.70

TP=3

FP=1

TN=10

FN=2

 $TPR = 3/(3+2) = 0.6$ $FPR = 1/(1+10) = 0.09$

2)

T=0.8

TP=1

FP=3

TN=12

FN=0

 $TPR = 1/1 = 1$ $FPR = 3/(3+12) = 0.2$

3)

T=0.9

TP=1

FP=3

TN=12

FN=0

 $TPR = 1/1$ $FPR = 3/(3+12) = 0.2$

the answer is T=0.7

Commento:

Domanda 10

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

1 point (-25% penalty for each wrong answer)

Which solutions can help address the challenge of variation in extracted requirements?

- ☐ a. Automatically rejecting extracted requirements with different phrasing
- ☐ b. Ignoring minor variations and focusing only on exact matches
- ☒ c. Using predefined synonym lists to map similar terms ✓
- ☒ d. Using word embeddings to detect semantically similar words ✓
- ☐ e. Manually reviewing every extracted requirement

Risposta corretta.

Le risposte corrette sono:

Using predefined synonym lists to map similar terms,

Using word embeddings to detect semantically similar words

Domanda 11

Risposta errata

Punteggio ottenuto -0,38 su 1,50

1.5 points (-25% penalty for a wrong answer)

The following reward can be used when fine-tuning a model on human feedback.

$$R(x, y) = r_{\theta}(x, y) - \beta \log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$$

Where r_{θ} is the output of a Reward Model, π_{new} and π_{old} are the new and the original policies, respectively (you can see them as the probabilities assigned by the new and original models).

What is the purpose of the term $\log \frac{\pi_{new}(y|x)}{\pi_{old}(y|x)}$?

- ☐ a. it increases the entropy of the output (i.e., it helps the model produce more "creative" answers)
- ☐ b. it improves the model's capability of following instructions
- ☐ c. it prevents the model from collapsing into a model that always predicts a single token
- ☐ d. it helps make the model more deterministic
- ☒ e. none of the other answers is correct ✗
- ☐ f. it prevents the new model from producing outputs that are too dissimilar from the original one
- ☐ g. it rewards the model when it generates an answer that humans will "like"

Your answer is incorrect.

La risposta corretta è:

it prevents the new model from producing outputs that are too dissimilar from the original one

1 point (-25% penalty for each wrong answer)

What is "priming" in the context of prompt engineering?

- ☐ a. Using reinforcement learning to fine-tune the model
- ☐ b. Pre-training the model with additional data

**Domanda
12**

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

- ☒ c. Providing initial input to guide the model's response ✓
- ☐ d. Adding noise to the input to test model robustness
- ☐ e. Resetting the model before every new prompt

Risposta corretta.

La risposta corretta è:

Providing initial input to guide the model's response